Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	i		Hypothesis 2, Because all three mutations only occur once. (In hypothesis 1 would need to occur twice)/ mutations occur just before the branch of mammals and dolphins/ OWTTE		1		1		
		ii		Compare amino acid {<u>sequence/ order</u>} in a (specific) protein (1) Greater similarity more closely related/ more recent common ancestor (1) OR Antibody antigen precipitation test(1) More precipitate the more related they are (1)	2			2		2
	(b)	i		Any three (x1)from: • HIV 1 has evolved from chimp (SIV) (1) • HIV2 has evolved from monkey (SIV) (1) • HIV1 and HIV 2 have evolved several times/ 5 strains of HIV (1) • 3 different origins for HIV1/ 2 different origins for HIV2 (1) • All forms of HIV from one common ancestor (1) must be correct context			3	3		
		ii		Predict when next mutation may occur	1			1		
		iii		SIV must have been evolving for a longer period/ ORA			1	1		
				Question 2 total	**3**	**1**	**4**	**8**	**0**	**2**

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